

A Deductive Engineering Approach to Subsurface Thermal Imaging: From 1D Diffusion to Mine Detection Algorithms

Gemini Thought Partner for Gujcost Robofest 2025

December 18, 2025

Abstract

This report provides a rigorous, step-by-step mathematical derivation of thermal contrast mechanisms. By mapping the thermodynamics of soil onto the complex impedance theory used in Electrical Engineering (EE), we deduce the "signature" of buried plastic mines, rocks, and surface clutter. This document is intended for engineers who require a deep understanding of the "how" and "why" behind thermal detection.

Contents

1	Introduction: The Problem of "Seeing" Under the Ground	2
2	Phase I: The Fundamental Thermodynamics	2
2.1	Fourier's Law (The Driver)	2
2.2	The Conservation of Energy (The Storage)	2
2.3	The 1D Heat Equation	2
3	Phase II: The Electrical Engineering Analogy	2
3.1	Complex Thermal Impedance	2
4	Phase III: Modeling the Buried Mine (Layered Media)	3
4.1	Reflection at the Interface	3
4.2	The Surface Contrast Equation	3
5	Phase IV: Surface Rocks vs. Buried Objects	3
5.1	Surface Exposed Rocks	3
6	Conclusion and SITL Implementation	3

1 Introduction: The Problem of "Seeing" Under the Ground

Autonomous drone-based mine detection relies on detecting anomalies in the ground's surface temperature. To a non-engineer, it is a mystery why a buried object affects the surface. To an engineer, the ground is a **spatial filter** and a **transmission line**. Our goal is to derive the transfer function of this system.

2 Phase I: The Fundamental Thermodynamics

Before we look at mines, we must define how heat moves through a solid.

2.1 Fourier's Law (The Driver)

Heat moves from high temperature to low temperature. We define the Heat Flux Vector $\vec{q}$ (W/m^2) as the "current" of thermal energy.

$$q = -k \frac{dT}{dz} \quad (1)$$

Intuition: If you have a temperature difference over a distance dz , the conductivity k determines how much energy flows. This is mathematically identical to Ohm's Law $J = \sigma E$.

2.2 The Conservation of Energy (The Storage)

Material has "Thermal Mass." If more heat flows *into* a small volume than flows *out*, the temperature must rise.

$$\rho c_p \frac{\partial T}{\partial t} = -\frac{\partial q}{\partial z} \quad (2)$$

Where ρ is density and c_p is specific heat. This is exactly how a capacitor stores charge ($i = C \frac{dV}{dt}$).

2.3 The 1D Heat Equation

Substituting (1) into (2):

$$\rho c_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) \implies \frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial z^2} \quad (3)$$

Where $\alpha = k/(\rho c_p)$ is **Thermal Diffusivity**. This equation describes how a "pulse" of heat spreads and weakens as it travels through a medium.

3 Phase II: The Electrical Engineering Analogy

For our project, the "signal" is the Sun. The Sun provides a periodic flux:

$$q_{in}(t) = Q_0 e^{j\omega t} \quad (4)$$

Where $\omega = 2\pi/86400$ rad/s (the 24-hour cycle).

3.1 Complex Thermal Impedance

Assuming the solution is $T(z, t) = \theta(z) e^{j\omega t}$, we plug this into (3):

$$j\omega \theta(z) = \alpha \frac{d^2 \theta}{dz^2} \implies \frac{d^2 \theta}{dz^2} - \gamma^2 \theta = 0 \quad (5)$$

Where $\gamma = \sqrt{j\omega/\alpha} = (1+j)\sqrt{\frac{\omega}{2\alpha}}$. The real part of γ is the **attenuation constant**. It defines the "Skin Depth" (δ) of the thermal wave:

$$\delta = \sqrt{\frac{2\alpha}{\omega}} \approx 10 - 15 \text{ cm for dry soil.} \quad (6)$$

The Thermal Impedance Z is defined as the ratio of Temperature to Flux:

$$Z = \frac{1}{\sqrt{j\omega k \rho c_p}} \quad (7)$$

Deduction: If the material has low conductivity and low density (like plastic), its Z is **high**. If it is dense and conductive (like rock), its Z is **low**.

4 Phase III: Modeling the Buried Mine (Layered Media)

A mine is not a point source; it is a boundary. Imagine a wave traveling down through the soil (Line 1) and hitting a plastic plate (Line 2).

4.1 Reflection at the Interface

Just like an EM wave hitting a load, some of the heat wave "bounces" back. The Reflection Coefficient Γ is:

$$\Gamma = \frac{Z_{mine} - Z_{soil}}{Z_{mine} + Z_{soil}} \quad (8)$$

- **For Plastic:** $Z_{mine} > Z_{soil} \implies \Gamma > 0$. The reflection is **in phase**. The reflected heat adds to the surface temperature. **Result: Hot Spot.**
- **For Rocks:** $Z_{rock} < Z_{soil} \implies \Gamma < 0$. The reflection is **out of phase**. It "sucks" heat away from the surface. **Result: Cold Spot.**

4.2 The Surface Contrast Equation

The total temperature you see with your drone camera is:

$$T_{surface} = T_{soil} (1 + \Gamma e^{-2\gamma L}) \quad (9)$$

Where L is the burial depth. The term $e^{-2\gamma L}$ shows that as L increases, the detection signal drops **exponentially**. This is why mines buried deeper than 15-20cm are virtually invisible.

5 Phase IV: Surface Rocks vs. Buried Objects

This is the most critical part for your AI algorithm to distinguish between a "real target" and "clutter."

5.1 Surface Exposed Rocks

A rock sitting **on** the ground is not part of the subsurface impedance model. It is a **direct solar absorber**.

$$Q_{abs} = \epsilon \cdot Q_{sun} - h(T_s - T_{air}) \quad (10)$$

Because the rock has a high absorption coefficient and high thermal mass, it absorbs energy all day without letting it dissipate into the air. At 3:00 PM:

- **Surface Rock:** Extremely hot (+15°C vs background). Sharp, clear edges.
- **Buried Mine:** Gently warm (+2°C to +4°C vs background). "Blurred" edges due to soil diffusion.

6 Conclusion and SITL Implementation

In your Gazebo/ArduPilot simulation, you must implement the following logic: 1. **Calculate Z for every voxel** based on its material. 2. **Apply the Harmonic Flux** at the top layer. 3. **Solve the 1D Impedance** for each vertical column to generate the 2D surface map. 4. **Detection Rule:** Flag circular anomalies where $\Delta T > 1.0^\circ\text{C}$ and the spatial gradient is "soft" (indicating burial).